

Network Testing Practical

Task 1: Test Communication Between Laptop & Phone (Ping Test)

- **What to do:**
 1. Connect your laptop and your friend's phone to the same Wi-Fi network.
 2. Find the phone's local IP address (go to the phone's **Wi-Fi Settings** → **Network Details**, e.g., 192.168.1.15).
 3. Open **Command Prompt** on Windows and run:

DOS

```
ping 192.168.1.15
```

- **Expected Result:** You should receive 4 successful replies: Reply from 192.168.1.15: bytes=32 time=... TTL=... (0% packet loss).

Task 2: Run Traceroute to YouTube

- **What to do:**
 - Open **Command Prompt** (or Terminal on Mac/Linux) and run:

DOS

```
tracert www.youtube.com
```

- **Deliverable:** Count and write down the total number of **hops** (routers passed) it takes to reach YouTube's destination server (typically between 8 to 14 hops).

Task 3: Disconnect Device & Record the Error Message

- **What to do:**

1. Turn off Wi-Fi on the phone (or disconnect it).
2. Re-run the ping command from your laptop:

DOS

ping 192.168.1.15

- **Deliverable:** Record the exact error message received:
 - Usually: Request timed out. or Destination host unreachable. (100% packet loss).

Task 4: Map Connected Devices Using the ARP Table

- **What to do:**
 - In Command Prompt, run:

DOS

arp -a

- **Deliverable:** Create a simple table identifying the devices on your Wi-Fi:

IP Address	MAC (Physical) Address	Assigned Device / Owner
192.168.1.1	xx-xx-xx-xx-xx-01	Home Wi-Fi Router (Gateway)
192.168.1.10	xx-xx-xx-xx-xx-2A	My Laptop
192.168.1.15	xx-xx-xx-xx-xx-B4	Friend's Android Smartphone

Task 5: One-Page Summary Report

- **What to include:**

1. **Network Setup:** Describe your home Wi-Fi environment (SSID, router model, subnet 192.168.1.0/24).
2. **Ping Results:** Compare the successful connected ping vs. the disconnected timeout test.
3. **Traceroute Findings:** Summarize the path to www.youtube.com (gateway →ISP nodes →Google backbone).
4. **Observations/Issues:** Mention any firewall blocking (such as Windows Defender ICMP drops) and attach command output screenshots.